H

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE NAME

BIOLOGY CLASS | REGISTRATION NUMBER

BIOLOGY 9648/03

Paper 3

15 September 2015

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in brackets [ ] at the end of each question or part question.

Section A
Answer all questions.
Write your answers in this booklet using dark blue or black pen.

Section B
Write your answers in this booklet using dark blue or black pen.
For Free Response Question, write your answers on the separate Answer Paper using dark blue or black pen.

For Examiner's Use	
1	/ 10
2	/ 10
3	/ 10
4	/ 10
5	/ 12
6	/ 20
TOTAL	**/ 72**

This document consists of **14** printed pages.

[Turn over

Section A

Answer <u>all</u> questions.

1 Fig. 1.1 shows the plasmid pACE15 which was isolated from bacteria and genetically engineered to contain the two selectable markers *tet*^{R} (confers cell resistance to antibiotic tetracyclin) and *lacZ* (encodes enzyme ß–galactosidase).

pACE15 is currently being used in biotechnology industries for the mass production of human growth hormone (hGH).In recombinant plasmids, the hGH gene is present within the Lac Z gene. Recombinant plasmids are subsequently transformed into *E. coli* cells. The bacterial cells from the transformation process are grown on an agar plate containing tetracycline and X-Gal.

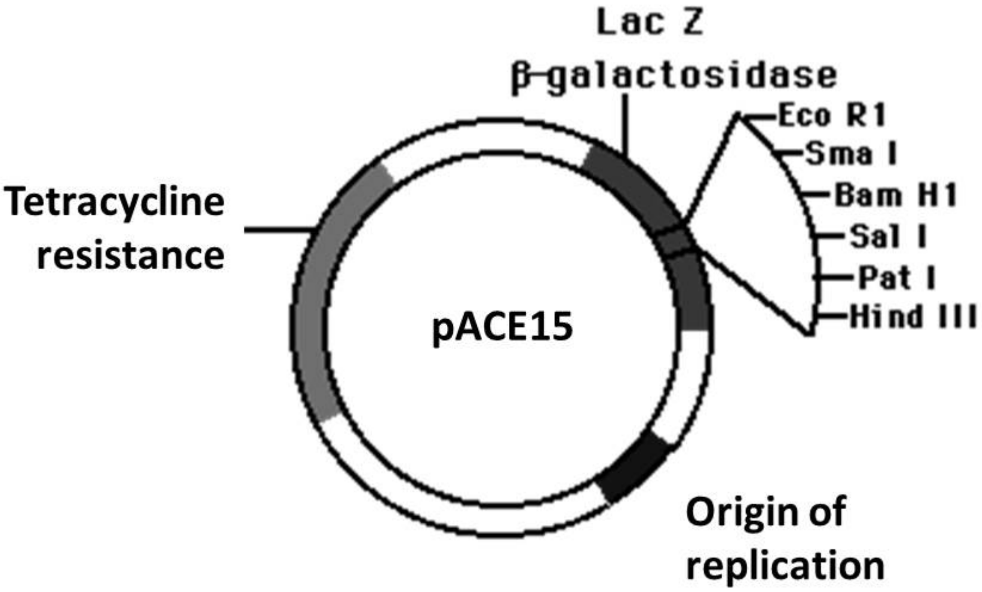

Fig. 1.1

(a) With reference to Figure 1.1, explain the feature of plasmid pACE15 that allows it to be used for gene cloning involving any species of organism. [2]

The presence of a multiple cloning site (MCS) containing many different and unique restriction sites that are recognised by a variety of restriction enzymes;

Allows DNA fragments to be isolated from any species of organism as restriction site cut by restriction enzyme used to isolate the DNA fragment should be found within the MCS (OWTTE);

OR

Origin of Replication; allows replication of plasmid including the inserted gene of interest, thereby resulting in many copies of gene of interest being produced.

OR

Selection marker e.g tetracyclin resistance;
That allows for selection of bacterial cells transformed with plasmid;

Table 1.1 shows two restriction enzymes that were considered for the process of inserting the hGH gene into the plasmid to form recombinant DNA.

Table 1.1

Restriction enzyme	**Restriction site**
EcoRI	↓ **5' GAATTC 3'** **3' CTTAAG 5'** ↑
SmaI	↓ **5' CCCGGG 3'** **3' GGGCCC 5'** ↑

(b) With reference to Table 1.1, state and explain which restriction enzyme would be more suitable for use in genetic engineering. [2]

EcoRI;

EcoRI generates sticky ends with a 5'AATT 3' and 3' TTAA 5' overhang which will allow complementary base pairing with overhangs of hGH gene cut with EcoRI, increasing the accuracy and efficiency of forming recombinant DNA compared to SmaI that produced blunt ends;

(c) With reference to Fig. 1.1, explain how transformed cells containing recombinant plasmids are differentiated from transformed cells containing re-annealed non-recombinant plasmids. [2]

In recombinant plasmid, the human DNA inserted disrupt lacZ gene / insertional inactivation and thus cannot produce functional β–galactosidase and X-gal in the culture medium is not cleaved / bacteria colonies would appear white;

Whereas in non-recombinant plasmid, the *lacZ* gene remains intact, able to code and produce functional β–galactosidase, so X-gal cleaved / bacteria colonies would appear blue;

(d) Describe how hGH gene in recombinant plasmids can be induced to undergo expression in *E.coli*. [1]

By adding lactose, which will induce expression of the hGH gene with Lac Z gene region;

(e) Suggest the advantage of using yeast cells instead of bacteria for the production of hGH protein. [2]

Since yeast is a eukaryote whereas bacteria are prokaryotes;

Allows post-translational modifications [eg. glycosylation] for proper folding/function of protein;

OR
Allows splicing of introns to produce functional protein so genomic DNA of hGH gene can be used/ Not necessary to perform reverse transcription and use cDNA of hGH gene;

[Total: 9]

2 Phenylketonuria (PKU) is an autosomal recessive inherited genetic disorder of metabolism where mutations in the PAH gene cause low levels of an enzyme called phenylalanine hydroxylase, that affects metabolism of phenylalanine. Without dietary treatment, phenylalanine can build up to harmful levels in the body, causing mental retardation and other serious problems.

A DNA segment, PK-21 found within 5 Centimorgans of the PAH gene has been identified. This segment exhibits restriction fragment length polymorphism (RFLP) that can be used as a marker for the genetic screening of PKU.

Fig. 2.1 shows the RFLPs of PK-21.

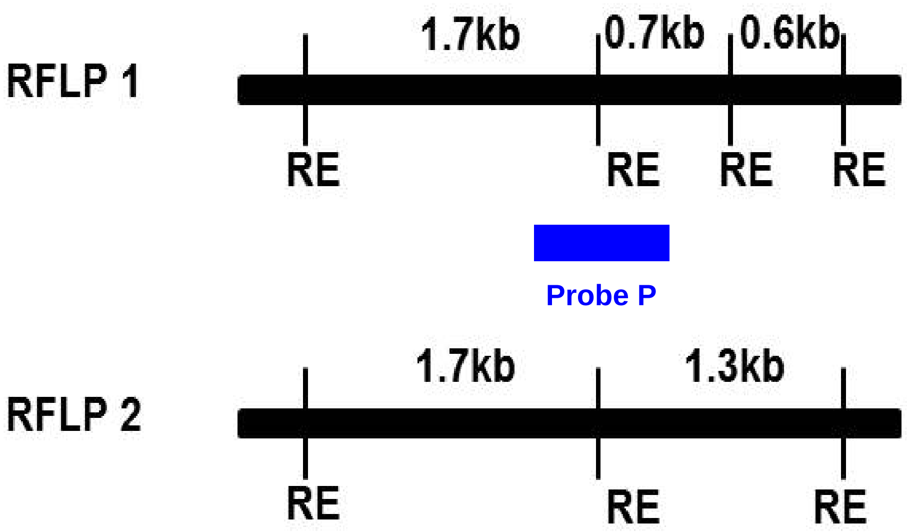

Fig. 2.1

Fig 2.2 shows the pedigree for the family and Fig 2.3 shows the genetic screening results for a family including that of the foetus the mother is carrying.

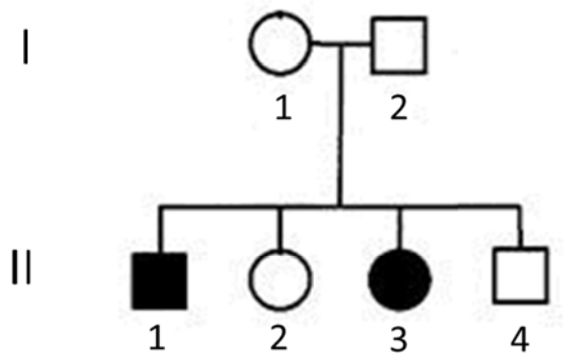

Fig. 2.2

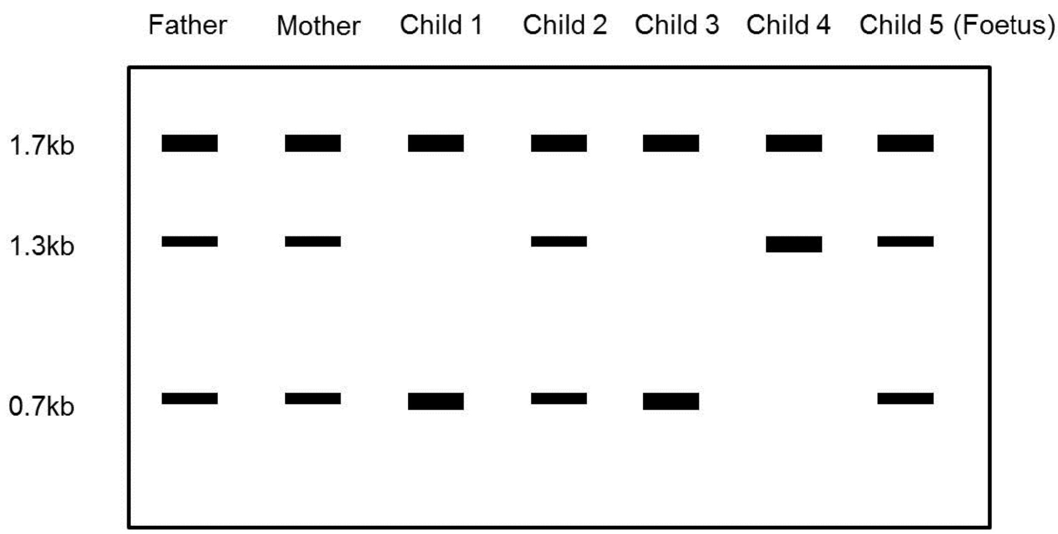

Fig. 2.3

(a) With reference to Fig. 2.1, explain one reason why PK-21 can be used as a genetic marker for the detection of the mutant PAH allele. [2]

Linkage
tightly linked to PAH gene / located very close on the same chromosome as PAH (within 5 centimorgans) / high chance that allele 1 and the mutant PAH allele would always be inherited together as one;

Polymorphic
Varying length of restriction fragments are generated upon cutting of the RFLP linked to normal allele (2 bands of lengths 1.7 and 1.3kb) and RFLP linked with the mutant PAH allele (3 bands of lengths 1.7, 0.7 and 0.6kb), allowing identification of the type of allele present in different individuals;

[Any one]

(b) The 2 RFLPs are detected using a probe P. With reference to Fig. 2.1 and 2.3, indicate on Fig 2.1, the position that probe P will bind. [1]

(c) Explain how gel electrophoresis is used to analyse and visualize DNA. [3]

Gel electrophoresis is a technique used to separate DNA fragments based on their molecular size where negatively charged DNA is loaded at the negative electrode and runs towards the positive electrode when an electric current is applied to it;

The gel acts as a molecular sieve and smaller sized fragments move further through the gel than larger fragments (Distance travelled inversely proportional to size of the fragment);

Ethidium bromide added binds to/ intercalates with DNA and observing the gel under UV light after the gel run will allow visualisation of bands/
The gel can be stained with methylene blue where the DNA bands will appear a darker blue and can be observed;

(d) With reference to Fig. 2.2 and 2.3, explain which RFLP the mutant PAH allele is linked to. [2]

RFLP 1;
Child 1 and 3, both affected by the disease, have 2 bands of 1.7 and 0.7kb, indicating that the 2 copies of the mutant PAH allele they possess are linked to RFLP 1;

(e) The newborn (Child 5) is diagnosed with PKU upon birth. Suggest a reason to explain the banding pattern observed for Child 5 in the genetic screening results. [1]

Though genetic test results suggest Child 5 is a carrier of PKU, the child is diagnosed with PKU, so crossing over must occurred between PK-21 and PAH loci during gamete formation in one of the parent, resulting in RFLP 1 was no longer linked to the mutant PAH allele;

[Total: 9]

3 Duchenne muscular dystrophy (DMD) is a genetic disease cuased by the absence of the protein dystrophin in muscle fibres. In the absence of dystrophin, muscle fibres gradually die.

A potential gene therapy for DMD involves injecting mucles with a viral vector carrying recombinant DNA (rDNA) for part of the normal allele for dystrophin.

(a) Outline the formation of recombinant DNA. [3]

1. **rDNA = DNA from two sources ;**
2. **both DNAs cut with, restriction enzyme / named restriction enzyme ;**
3. **giving sticky ends ; or giving blunt ends to which sticky ends added ;**
4. **complementary binding of sticky ends ;**
5. **H bonds / e.g. A to T / e.g. C to G ;**
6. **nicks in (sugar-phosphate) backbone sealed by ligase ;**

(b) Mice with the symptoms of DMD were given this gene therapy shortly after birth. Each mouse was injected with the viral vector in a muscle of one hind limb. The corresponding muscle of the other hind limb was injected with a buffer solution to provide a control. The nuclei of muscle fibres that do not produce dystrophin move from the edge of the fibre to the centre. The fibres eventually die. The percentage of muscle fibres with centrally placed nuclei was measured in fibres from treated and control muscles at different times after injection. The results are shown in the figure below.

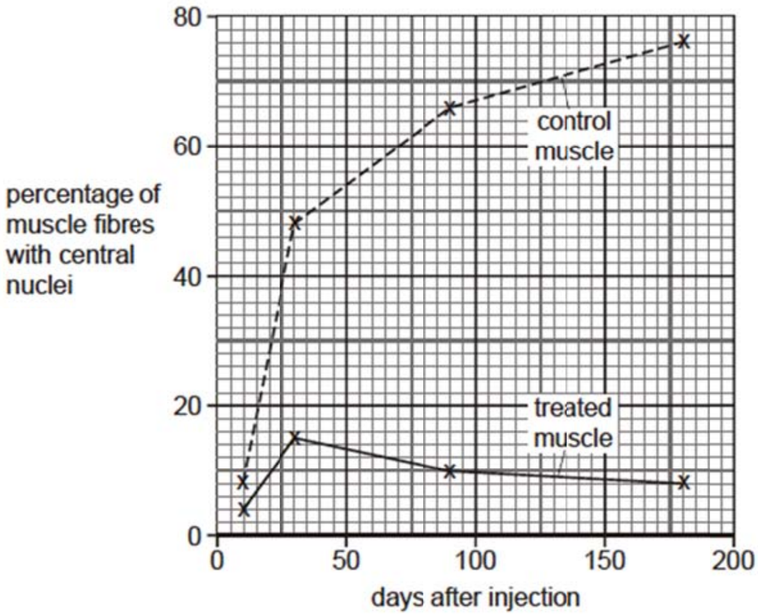

Fig. 3.1

(i) Identify the type of gene therapy treatment stated above [1]

1. **In-vivo viral gene therapy**

(ii) Using the information above, describe the results of the experiment. [3]

1. **percentage / proportion, of, muscle fibres with central nuclei / dying muscle fibres, increases in control with time ;**
2. **percentage / proportion, of, muscle fibres with central nuclei / dying muscle fibres, reduced by treatment ;**
3. **ref to comparative figures with percentages and day ;**

(iii) Explain why it is easier to perform gene therapy when the normal allele is the dominant allele of the gene concerned. [2]

1. **Has effect when added to the genome;**
2. **Not masked;**
3. **No need to remove / inactivate the recessive / mutant allele;**

(iv) Genetic screening is available for families with a history of DMD. Discuss the advantages and disadvantages of genetic screening. [4]

Advantages

1 can identify presence of disorder ;
2 removes uncertainty ;
3 allows early treatment ;
4 which may improve, life expectancy / quality of life ; A avoid unnecessary suffering 5 allows, informed choice about having children / planning healthy family ;
6 allows IVF and, embryo screening / preimplantation genetic diagnosis (PGD) ;
7 allows fetal testing and termination ;
8 choice, re donation / adoption ;
9 AVP ; e.g. detail of donation: AI(D) / egg donation / embryo donation
[any 2]

Disadvantages

10 false, positives / negatives ;
11 may not be test for all mutations ;
12 only small number tests available / not available for all conditions ;
13 simple presence may not result in condition ;
14 confirmed presence gives stress / fear ;
15 problem re, telling / testing, rest of family ;
16 discrimination by, employers / insurers ;
17 ethics of termination ;
18 AVP ; e.g. detail of problem of test, risk of test procedure, diagnosis and elimination rather than treatment, increase in, intolerance / discrimination, of disabled, 'designer' problem
[any 2]

[Total: 13]

Name: ________________________ Class: 2bi2___ / 2IPbi2___

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Section B

Answer **all** the questions in this section.

4 Experimental results below show the vulnerability of using various Orchid plant parts in carrying out micropropagation, at various times of the year.

Table 4.1

explant	time of year	number of explants	number of cultured explants without any microbe contamination	Percentage of cultured explants without microbe contamination/ %
leaf disc	May	153	12	**8**
	September	322	16	**5**
	February	332	30	**9**
stem tissue	May	194	116	**60**
	September	191	122	**64**
	February	211	156	**74**

(a) With reference to **Table 4.1**, compare the effect of the different methods of taking tissue samples and the time of year on microbe contamination of the cultured explants. [3]

stem tissue has less contamination than leaf discs for all three times of the year;
for example, in May, 60% of stem tissue has no fungal or bacterial contamination compared to only 8% of leaf discs or other appropriate comparison with manipulated data;
both treatments have highest % non-contamination in Feb of 74% for stem tissue and 9% for leaf discs respectively;
lowest for leaf discs in Sep of 5%, for stem tissue in May of 8%;

(b) Suggest why it is crucial for the removal of pathogens from the explants. [2]

May compete with the callus culture/plant cells for nutrients;

Growth/development may be stunted (OTTWE);

(c) Explain the advantages of using cloned plants in the farming of orchids. [3]

Large numbers of genetically identical plants can be easily obtained;
Plantlets / callus are compact, hence easily kept in identical conditions;
Can be obtained all year round independent of growing seasons ;

(d) State one concern of farming genetically modified herbicide resistant corn and suggest a way of overcoming it. [2]

Resistance genes may be passed onto <u>weed relatives</u> through <u>cross-pollination/ horizontal gene transfer;</u>
[Reject: gene transfer]

Create a large <u>buffer zone</u> / isolation distance by planting <u>unrelated crops</u>;

OR

Intense use of same herbicide will eventually select for resistant weeds, for which gmo crops will no longer be effective;

Alternate use of gmo crops with other types of pesticide;

OR

Forced dependence of farmers on gmo corns as they are not allowed to replant seeds from their crops (WTTE);

Allow farmers to replant seeds;

OR

Allows excessive use of herbicide, causing runoffs which are toxic to fish and amphibians;

Promote responsible use of herbicide among farmers;

OR

Any logical concern without solution, max 1

[Total: 10]

5 Planning Question

Write your answers to this question on the separate answer paper provided.

You are required to plan an experiment to determine the effects of the growth regulator gibberellin (GA) on the germination of maize grains.

You are required to use the following information to assist in your experiment plan:

- Maize needs to be soaked in water for 24 hours to stimulate germination
- Maize starts to germinate 48 – 72 hours after soaking
- GA promotes germination by activating a gene that causes the synthesis of the enzyme amylase
- The range of concentrations at which GA is found in plants is between 1 – 10 µmol dm-3
- Amylase is released by the aleurone layer into the endosperm of the maize grain to hydrolyse the starch reserves
- The activity of amylase can be estimated by cutting the maize grains lengthways into two halves and placing the cut sides onto agar containing Starch in Petri dishes
- After incubation at a constant temperature, iodine solution is used to test the agar for the presence of starch

Your planning must be based on the assumption that you have also been provided with the following materials which you must use:

- 3 mmol dm-3 GA solution

In addition, you may include in your plan the usage of any equipment that can be found in a standard school laboratory.

Your plan should have a clear and helpful structure to include:

- an explanation of the theory to support your practical procedure.
- a description of the method used, including the scientific reasoning behind the method
- an explanation of the dependent, independent variables and controlled variables
- how you will record your results and ensure that they are as accurate & reliable as possible.
- proposed layout of results tables and graph with clear headings and labels
- recommended safety measure.
- the correct use of technical and scientific terms.

[Total: 12]

Parts	Expected Answer	Marks	Additional Guidance
Introduction	**Explanation of theory:** (1) Effect of amylase on agar containing starch (2) Effect of GA on amylase concentration **Brief Methods:** (1) Determine the effect of GA by determining the diameter of brown coloration around the seeds **Hypothesis:** (1) Higher the GA concentration, larger the diameter of brown coloration around the seeds	[1] [1] [1]	
Variables	**Independent:** Concentration of GA **Dependent:** Ref. to starch free area / zone (around grains) determined by the diameter of brown coloration **Controlled Variable:** (1) Using same / stated number of grains / halves, for each concentration (2) Stated and same volume of each soaking solution / GA (3) Method of keeping same incubation temperature (4) Time for incubation of enzyme (5) Concentration of starch in agar (6) Same volume / depth of agar in petri dishes (7) Covering to prevent evaporation / contamination	[1] [1]	R: amount / quantity R: amylase activity

Methodology	(1) Ref: to method of diluting the (3 mmol. dm^{-3}) GA to give a minimum of (any) 5 dilutions; (2) Ref: to concentration (other than 0) that fall in the range of 3 mmol.dm^{-3} to any value above zero with units (3) Ref: to soaking grains in GA solutions for, minimum 24 hours and maximum 72 hours (4) Ref: to a control using (distilled/deionized) water; (5) Ref: to stated incubation temperature (6) Ref: to a suitable method of measuring the starch free area around the grains; (7) Ref: to presence of repeats and replicates	[4] Each point 1 mark	1. Series / serial / simple dilution as method. Dilution table should be provided 2. Minimum is 2 values that are not higher than 3 mmol. dm^{-3} and are above zero 3. Ignore if the range of GA is incorporated into agar plates. 4. Accept description – "to allow comparison " 5. E.g. any one temperature in the range of 15 – 35°C. Reject body temperature (37°C) 6. E.g. trace outline onto a grid and use the grid to count the number of squares / use suitable apparatus e.g. calipers to measure radius or diameter
Data	(1) Table of results with suitable headings depending on the methods used (must show considerations of repeats in the table) (2) Graph with possible trend (x-axis: concentration of GA and mmol. dm^{-3} ; y-axis: (mean) area of clear zone and mm^2)	[1] [1]	

Safety Caution	**(1) E.g Cutting away from hands / using tiles for cutting** **(2) E.g. ref. to allergy / irritation → use of gloves / mask / eye protection**	**[1]**	

Free Response Question

Write your answers to this question on the separate answer paper provided.
Begin each section of the question on a new piece of answer paper.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b), etc., as indicated in the question.

6 **(a)** Discuss the factors which prevent gene therapy from becoming an effective treatment. [8]

a. Therapy is typically <u>short-lived</u> because of <u>difficulty in integrating</u> normal, functional allele into host cell genome
b. <u>natural death of treated cells</u> means that new cells need to be constantly transformed;
c. Many diseases are a result of <u>multiple genes and/or multiple factors</u> (e.g. Alzheimer's disease, diabetes etc)
OR
as such will be difficult to treat with gene therapy which currently treats only <u>single gene disorders</u>;
d. Some diseases such as <u>thalassemia</u> require a <u>precise level of expression</u> of the normal gene, which gene therapy cannot achieve [with reason]
e. For viral mediated gene therapy, problems associated with the viral vector include immune response against the vector such as allergies, inflammatory and toxicity responses [give one example];
f. Cannot find a suitable viral vector due to viruses' <u>host specificity</u>
g. Viruses may <u>recover its ability to cause disease</u> once in host;
h. If virus vector triggers the immune response the first time, the immune system will mount a stronger and faster response the second time rendering the therapy <u>ineffective subsequently</u>;
i. <u>Insertional mutagenesis</u> caused by retroviruses may cause inactivation of tumour suppressor genes and/or activation of proto-oncogenes, leading potentially to cancer;

(b) Using blood stem cells as example, describe the unique features and functions of stem cells.

Explain the differences between stem cells and cancer cells. [12]

Features: (max 3)

a. Unspecialised/undifferentiated cells with no tissue specific structures/do not carry out any specific function;
b. Can undergo mitosis/divide indefinitely and undergo self-renewal to renew pool of stem cells;
c. They have the ability to differentiate to produce specialized cells upon receiving appropriate molecular signals;

Blood stem cells (max 6)

1 multipotent
2 able to differentiate into a limited range of cell type / produce only cells of a specific lineage
3 daughter cells are not pluripotent or totipotent
4 can differentiate into RBC, WBC, platelets
5 Function = replacement of worn out blood cells / daily blood turnover and fighting infections
6 lost through normal wear and tear, disease, injury

Differences: (max 5)

d. Unlike stem cells, cancer cells do not differentiate and is unresponsive to molecular signals;
e. Cancer cells divide indefinitely while stem cell division is determined by molecular signals [that either stimulates cell division or stop it altogether];
f. Cancer cells experience no contact inhibition and is invasive while stem cells experience contact inhibition;
g. Cancer cells metastasize (dislodge from original tumour and form secondary tumours) while stem cells remain in tissue of origin;
h. AVP of comparison;

[Total: 20]

--- End of Paper ---